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Inventor(s)	Eng; Adam et al.

Methods and systems for providing variable bitrate content

Abstract

Provided are methods and systems for providing variable bitrate content (e.g., video content, audio content, multimedia content, etc.). The content may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). Each portion of the content may be tagged and/or associated with a content element. A content element may be associated with and/or indicate one or more of an encoding parameter associated with a respective portion of the content, attributes of the content (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content, or additional content related items (e.g., one or more advertisements, etc.) associated with the respective portion of the content. The content element may be used to determine a bitrate to associate with the respective portion of the content. The content may be received/retrieved according to the determined bitrates.

Inventors:	Eng; Adam (Golden, CO), Eng; David (Louisville, CO)
Applicant:	Comcast Cable Communications, LLC (Philadelphia, PA)
Family ID:	1000008820932
Assignee:	Comcast Cable Communications, LLC (Philadelphia, PA)
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Primary Examiner: Zhou; Zhihan

Attorney, Agent or Firm: Ballard Spahr LLP

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION (1) This application is a continuation of U.S. application Ser. No. 16/265,804, filed on Feb. 1, 2019, the entirety of which is incorporated by reference herein.

BACKGROUND

(1) Adaptive bitrate systems provide content that transitions between bitrates based on network conditions such as congestion, errors, or the like. A change in bitrate associated with the content may occur at a point in the content where a quality of the content may be negatively impacted by the change in bitrate, such as during an important scene or during a scene transition. A negative impact may include lost video frames, dropped audio, misaligned audio and video, unintentional pauses, and other undesirable effects.

SUMMARY

(2) It is to be understood that both the following general description and the following detailed description are examples and explanatory only and are not restrictive. Methods and systems for providing variable bitrate content are described.

(3) Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). Each portion of the content may be tagged and/or associated with a content element. For example, each portion of the content may be tagged and/or associated with a content element that indicates an encoding complexity associated with the respective portion of the content. Further, a content element may be associated with and/or indicate one or more of metadata associated with the respective portion of the content, attributes of the content (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content, additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with the respective portion of the content, or the like. The content element, based on a user viewing (e.g., accessing, consuming, etc.) the respective portion of the content, may be used to determine a bitrate to associate with the respective portion of the content. The user, based on the content element, may view the respective portion of the content at the bitrate. The content may be transmitted to a network device (e.g., a content player, a media device, a smart device, a mobile device, a computing device, etc.) such that each portion of the content is provided at a bitrate determined (e.g., variable bitrate) based on the content element associated with the respective portion of the content.

(4) This summary is not intended to identify critical or essential features of the disclosure, but merely to summarize certain features and variations thereof. Other details and features will be described in the sections that follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, provide examples and together with the description, serve to explain the principles of the methods and systems:

(2) FIG. 1 shows a system for variable bitrate content;

(3) FIG. 2 shows a diagram for variable bitrate content;

(4) FIG. 3 shows a flowchart of a method for variable bitrate content;

(5) FIG. 4 shows a flowchart of a method for variable bitrate content;

(6) FIG. 5 shows a flowchart of a method for variable bitrate content;

(7) FIG. 6 shows a flowchart of a method for variable bitrate content;

(8) FIG. 7 shows a flowchart of a method for variable bitrate content; and

(9) FIG. 8 shows a block diagram of a computing device for implementing variable bitrate content.

DETAILED DESCRIPTION

(10) As used in the specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such

a range is expressed, another configuration includes from the one particular value and/or to the other particular value. When values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another configuration. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

(11) “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes cases where said event or circumstance occurs and cases where it does not.

(12) Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude other components, integers or steps. “Exemplary” means “an example of” and is not intended to convey an indication of a preferred or ideal configuration. “Such as” is not used in a restrictive sense, but for explanatory purposes.

(13) It is understood that when combinations, subsets, interactions, groups, etc. of components are described that, while specific reference of each various individual and collective combinations and permutations of these may not be explicitly described, each is specifically contemplated and described herein. This applies to all parts of this application including, but not limited to, steps in described methods. Thus, if there are a variety of additional steps that may be performed it is understood that each of these additional steps may be performed with any specific configuration or combination of configurations of the described methods.

(14) As will be appreciated by one skilled in the art, hardware, software, or a combination of software and hardware may be implemented. Furthermore, a computer program product on a computer-readable storage medium (e.g., non-transitory) having processor-executable instructions (e.g., computer software) embodied in the storage medium. Any suitable computer-readable storage medium may be utilized including hard disks, CD-ROMs, optical storage devices, magnetic storage devices, memristors, Non-Volatile Random Access Memory (NVRAM), flash memory, or a combination thereof.

(15) Throughout this application reference is made to block diagrams and flowcharts. It will be understood that each block of the block diagrams and flowcharts, and combinations of blocks in the block diagrams and flowcharts, respectively, may be implemented by processor-executable instructions. These processor-executable instructions may be loaded onto a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the processor-executable instructions which execute on the computer or other programmable data processing apparatus create a device for implementing the functions specified in the flowchart block or blocks.

(16) These processor-executable instructions may also be stored in a computer-readable memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the processor-executable instructions stored in the computer-readable memory produce an article of manufacture including processor-executable instructions for implementing the function specified in the flowchart block or blocks. The processor-executable instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the processor-executable instructions that execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

(17) Accordingly, blocks of the block diagrams and flowcharts support combinations of devices for performing the specified functions, combinations of steps for performing the specified functions and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flowcharts, and combinations of blocks in the block diagrams and flowcharts, may be implemented by special purpose hardware-based computer

systems that perform the specified functions or steps, or combinations of special purpose hardware and computer instructions.

(18) “Content items,” as the phrase is used herein, may also be referred to as “content,” “content data,” “content information,” “content asset,” “multimedia asset data file,” or simply “data” or “information”. Content items may be any information or data that may be licensed to one or more individuals (or other entities, such as business or group). Content may be electronic representations of video, audio, text and/or graphics, which may be but is not limited to electronic representations of videos, movies, or other multimedia, which may be but is not limited to data files adhering to MPEG2, MPEG, MPEG4 UHD, HDR, 4k, Adobe® Flash® Video (.FLV) format or some other video file format whether such format is presently known or developed in the future. The content items described herein may be electronic representations of music, spoken words, or other audio, which may be but is not limited to data files adhering to the MPEG-1 Audio Layer 3 (.MP3) format, Adobe®, CableLabs 1.0, 1.1, 3.0, AVC, HEVC, H.264, Nielsen watermarks, V-chip data and Secondary Audio Programs (SAP). Sound Document (.ASND) format or some other format configured to store electronic audio whether such format is presently known or developed in the future. In some cases, content may be data files adhering to the following formats: Portable Document Format (.PDF), Electronic Publication (.EPUB) format created by the International Digital Publishing Forum (IDPF), JPEG (.JPG) format, Portable Network Graphics (.PNG) format, dynamic ad insertion data (.csv), Adobe® Photoshop® (.PSD) format or some other format for electronically storing text, graphics and/or other information whether such format is presently known or developed in the future. Content items may be any combination of the above-described formats.

(19) “Consuming content” or the “consumption of content,” as those phrases are used herein, may also be referred to as “accessing” content, “providing” content, “viewing” content, “listening” to content, “rendering” content, or “playing” content, among other things. In some cases, the particular term utilized may be dependent on the context in which it is used. Consuming video may also be referred to as viewing or playing the video. Consuming audio may also be referred to as listening to or playing the audio.

(20) This detailed description may refer to a given entity performing some action. It should be understood that this language may in some cases mean that a system (e.g., a computer) owned and/or controlled by the given entity is actually performing the action.

(21) Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). Each portion of the content may be tagged and/or associated with a content element. A content element may be associated with an encoding parameter associated with the respective portion of the content. The encoding parameter may indicate a level of complexity (e.g., low complexity, medium complexity, high complexity, etc.) to encode the respective portion of the content. The level of complexity may be based on one or more of a prediction frame error associated with the respective portion of the content, a relative size of an inter-coded frame of the respective portion of the content with respect to an intra-coded frame of the respective portion of the content, or a number of motion vectors to encode a frame of the respective portion of the content. The level of complexity may be based on and/or determined by an amount of change between a plurality of pixels associated with a previous portion of the content and a corresponding plurality of pixels associated with the respective portion of the content. For example, a level of complexity to encode a portion of content may be low if an amount of change in pixels between the portions of content does not satisfy a threshold, or the level of complexity to encode the portion of content may be high if the amount of change in pixels between the portions of content satisfies or exceeds the threshold.

(22) The content element may be associated with attributes of content (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content. For example, the content element may comprise and/or be associated with metadata that indicates and/or causes a scene

transition between a previous portion (or a later portion) of content and the respective portion of the content. As another example, the content element may comprise and/or be associated with metadata that associates the respective portion of the content with one or more advertisements. As another example, the content element may comprise and/or be associated with metadata that associates an audio level with the respective portion of the content.

(23) Each portion of the content (e.g., video content, audio content, multimedia content, etc.) may be tagged and/or associated with a content element that is used to determine (e.g., vary, adapt, etc.) a bitrate for the respective portion of content. The content may be provided to a network device (e.g., a content player, a media device, a smart device, a mobile device, a computing device, etc.) such that each portion of the content is provided at a bitrate determined based on the content element associated with the respective portion of the content.

(24) FIG. 1 shows an example system in which the present methods and systems may operate. Those skilled in the art will appreciate that present methods may be used in systems that employ both digital and analog equipment. One skilled in the art will appreciate that provided herein is a functional description and that the respective functions may be performed by software, hardware, or a combination of software and hardware.

(25) A system **100** may comprise a central location **101** (e.g., a headend, HUB, or a Data Center), which may receive content (e.g., data, input programming, and the like) from multiple sources. The central location **101** may combine the content from the various sources and may distribute the content to user (e.g., subscriber) locations (e.g., location **119**) via a network **116**.

(26) The central location **101** may receive content from a variety of sources **102a**, **102b**, **102c**. The content may be transmitted from the source to the central location **101** via a variety of transmission paths, including wireless (e.g. satellite paths **103a**, **103b**) and a terrestrial path **104**. The central location **101** may also receive content from a direct feed source **106** via a direct line **105**. Other input sources may comprise capture devices such as a video camera **109** or a server **110**. The signals provided by the content sources may include a single content item or a multiplex that includes several content items.

(27) The central location **101** may comprise one or a plurality of receivers **111a**, **111b**, **111c**, **111d** that are each associated with an input source. For example, MPEG encoders such as an encoder **112**, are included for encoding local content or a video camera **109** feed. A switch **113** may provide access to the server **110**, which may be a Pay-Per-View server, a data server, an internet router, a network system, a phone system, and the like. Some signals may require additional processing, such as signal multiplexing, prior to being modulated. Such multiplexing may be performed by a multiplexer (mux) **114**.

(28) The central location **101** may comprise one or a plurality of modulators **115** for interfacing to the network **116**. The modulators **115** may convert the received content into a modulated output signal suitable for transmission over a network **116**. The output signals from the modulators **115** may be combined, using equipment such as a combiner **117**, for input into the network **116**. The network **116** may comprise a content delivery network, a content access network, and/or the like. For example, the network **116** may provide content from a variety of sources using a variety of network paths, protocols, devices, and/or the like. The content delivery network and/or content access network may be managed (e.g., deployed, serviced) by a content provider, a service provider, and/or the like.

(29) A control system **118** may permit a system operator to control and monitor the functions and performance of the system **100**. The control system **118** may interface, monitor, and/or control a variety of functions, including, but not limited to, the channel lineup for the television system, billing for each user, conditional access for content distributed to users, and the like. The control system **118** may provide input to the modulators for setting operating parameters, such as system specific MPEG table packet organization or conditional access information. The control system **118** may be located at the central location **101** or at a remote location.

- (30) The network **116** may distribute signals from the central location **101** to user locations, such as a user location **119**. The network **116** may comprise an optical fiber network, a coaxial cable network, a hybrid fiber-coaxial network, a wireless network, a satellite system, a direct broadcast system, an Ethernet network, a high-definition multimedia interface network, universal serial bus network, or any combination thereof.
- (31) The network **116** may comprise a network component **129**. The network component **129** may comprise any device, module, and/or the like communicatively coupled to the network **116**. For example, the network component **129** may comprise a router, a switch, a splitter, a packager, a gateway, an encoder, a storage device, a multiplexer, a network access location (e.g., tap), physical link, and/or the like.
- (32) A multitude of users may be connected to the network **116** at one or more of the user locations. At the user location **119**, a media device **120** may demodulate and/or decode, if needed, the signals for display on a display device **121**, such as on a network device, a television set (TV), or a computer monitor. For example, the media device **120** may comprise a demodulator, decoder, frequency tuner, and/or the like. The media device **120** may be directly connected to the network (e.g., for communications via in-band and/or out-of-band signals of a content delivery network) and/or connected to the network **116** via a communication terminal **122** (e.g., for communications via a packet switched network). The media device **120** may comprise a network device, a set-top box, a digital streaming device, a gaming device, a media storage device, a digital recording device, a combination thereof, and/or the like. The media device **120** may comprise one or more applications, such as content viewers, social media applications, news applications, gaming applications, content stores, electronic program guides, and/or the like. Those skilled in the art will appreciate that the signal may be demodulated and/or decoded in a variety of equipment, including the communication terminal **122**, a computer, a TV, a monitor, or satellite dish.
- (33) The communication terminal **122** may be located at the user location **119**. The communication terminal **122** may communicate with the network **116**. The communications terminal **122** may comprise a network device, a modem (e.g., cable modem), a router, a gateway, a switch, a network terminal (e.g., optical network unit), and/or the like. The communications terminal **122** may be configured for communication with the network **116** via a variety of protocols, such as internet protocol, DOCSIS protocol, transmission control protocol, file transfer protocol, session initiation protocol, voice over internet protocol, and/or the like. For example, for a cable network, the communication terminal **122** may provide network access via a variety of communication protocols and standards, such as Data Over Cable Service Interface Specification.
- (34) The user location **119** may comprise a first access point **123**, such as a wireless access point. The first access point **123** may provide one or more wireless networks in at least a portion of the user location **119**. The first access point **123** may provide access to the network **116** to devices configured with a compatible wireless radio, such as a mobile device **124**, the media device **120**, the display device **121**, or other computing devices (e.g., laptops, sensor devices, security devices). For example, the first access point **123** may provide a user managed network (e.g., local area network), a service provider managed network (e.g., public network for users of the service provider), and/or the like. It should be noted that in some configurations, some or all of the first access point **123**, the communication terminal **122**, the media device **120**, and the display device **121** may be implemented as a single device.
- (35) The user location **119** may not be fixed. By way of example, a user may receive content from the network **116** on the mobile device **124**. The mobile device **124** may comprise a network device, laptop computer, a tablet device, a computer station, a personal data assistant (PDA), a smart device (e.g., smart phone, smart speaker, smart apparel, smart watch, smart glasses), GPS, a vehicle entertainment system, a portable media player, a combination thereof, and/or the like. The mobile device **124** may communicate with a variety of access points (e.g., at different times and locations or simultaneously if within range of multiple access points). For example, the mobile device **124**

may communicate with a second access point **125**. The second access point **125** may be a cell tower, a wireless hotspot, another mobile device, and/or other remote access point. The second access point **125** may be within range of the user location **119** or remote from the user location **119**. For example, the second access point **125** may be located along a travel route, within a business or residence, or other useful locations (e.g., travel stop, city center, park).

(36) The system **100** may comprise an application device **126**. The application device **126** may be a computing device, such as a server. The application device **126** may provide services related to applications. For example, the application device **126** may comprise an application store. The application store may allow users to purchase, download, install, upgrade, and/or otherwise manage applications. For example, the application device **126** may allow users to download applications to a device, such as the mobile device **124**, communications terminal **122**, the media device **120**, the display device **121**, and/or the like. The application device **126** may run one or more application services to provide data, handle requests, and/or otherwise facilitate operation of applications for the user.

(37) The system **100** may comprise one or more content source(s) **127**. The content source(s) **127** may provide content (e.g., video, audio, games, applications, data) to the user. The content source(s) **127** may provide streaming media, such as on-demand content (e.g., video on-demand), content recordings, and/or the like. For example, the content source(s) **127** may be managed by third party content providers, service providers, online content providers, over-the-top content providers, and/or the like. The content may be provided via a subscription, by individual item purchase or rental, and/or the like. The content source(s) **127** may provide the content via a packet switched network path, such as via an internet protocol (IP) based connection. The content may be accessed by users via applications, such as mobile applications, television applications, set-top box applications, gaming device applications, and/or the like. An example application may be a custom application (e.g., by content provider, for a specific device), a general content browser (e.g., web browser), an electronic program guide, and/or the like.

(38) The system **100** may comprise a content management device **128**. The content management device **128** may provide content, services, and/or the like to a network device. For example, the content management device **128** may be one of a plurality of content management devices distributed across the network **116**. The content management device **128** may be located in a region proximate to the user location **119**. A request for content from a network device may be directed to the content management device **128** (e.g., due to the location of the content management device and/or network conditions).

(39) The content management device **128** may package content (e.g., video content, audio content, multimedia content, etc.) for delivery to a network device (e.g., in a specific format requested by the network device), provide the network device a manifest file (e.g., or other index file describing portions of the content), provide streaming content (e.g., unicast, multicast), provide a file transfer, and/or the like. To enable faster delivery of content to network devices, the content management device **128** may cache or otherwise store content (e.g., frequently requested content). To enable quality in delivery of content, the content management device **128** may associate content with a content element. The content management device **128** may associate content and/or portions of content to a bitrate or a plurality of bitrates according to a content element or a plurality of content elements.

(40) Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). The content management device **128** may tag and/or associate each portion of the plurality of portions of the content with a content element of a plurality of content elements. As previously mentioned and later described, a content element may be used to determine a bitrate to associate with a portion of content.

(41) A content element may comprise and/or be associated with an encoding parameter. The encoding parameter may be associated with a portion of a plurality of portions of content. The

content management device **128** may comprise an encoding module **130**. The encoding module **130** may encode content (e.g., video content, audio content, multimedia content, etc.), such as content provided by the content source(s) **127**. The encoding module **130** may comprise one or more encoders configured to receive content (e.g., receive content from the content source(s) **127**, etc.) and encode the content. For example, the encoding module **130** may encode each portion of the plurality of portions of the content. As later described, each portion of the plurality of portions of the content may be associated with a bitrate and/or a plurality of bitrates. For example, each portion of the plurality of portions of the content may be associated with a bitrate and/or a plurality of bitrates based on an encoding parameter (e.g., a content element) associated with the respective portion of the content. The encoding module **130** may encode the content into a compressed and/or encrypted format. For example, the encoding module **130** may encode the content into an MPEG stream.

(42) As previously described, content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.) and the encoding module **130** may encode each portion of the plurality of portions of the content. For example, the encoding module **130** may perform intra-frame and inter-frame encoding (e.g., compression). Intra-frame encoding may comprise encoding a frame of content (e.g., a frame of video) by reference to the frame itself. Inter-frame encoding may comprise compressing a frame a plurality of frames of content, by reference to one or more other frames of the plurality of frames. For example, an intra-coded frame (“I-frame”) may comprise a frame of content that is encoded without reference to other frames. A prediction coded frame (“P-frame”) may comprise a frame of content encoded with reference to another frame, such as an I-frame. A bi-directionally prediction coded (“B-frame”) frame may comprise a frame of content encoded with reference to multiple frames. For example, the encoding module **130** may encode a content stream into a plurality of I-frames, P-frames, and B-frames. The plurality of I-frames, P-frames, and B-frames may be organized into groups, each group known as a group of frames and/or group of pictures (GOP).

(43) To encode a frame of content (e.g., video content, audio content, multimedia content, etc.) with reference to another frame, the encoding module **130** may determine/encode one or more motion vectors configured to correlate a portion of the encoded frame to a portion of a referenced frame. The one or more motion vectors may indicate a difference in location between one or more pixels of the encoded frame and one or more corresponding (e.g., identical, similar, etc.) pixels in the reference frame. A motion vector may comprise, for example, a direction and distance between two points in a coordinate system. As another example, a motion vector may comprise a coordinate in a reference frame and a coordinate in the encoded frame. The encoding module **130** may encode an I-frame by encoding all the pixels in a frame. The encoding module **130** may encode P-frames and/or B-frames without encoding all of the pixels in a frame. The encoding module **130** may encode motion vectors that associate (e.g., correlate) portions (e.g., pixels) of a reference frame and the location thereof with portions of an encoded frame and the location thereof. The encoding module **130** may determine prediction errors associated with encoding a frame and/or a plurality of frames. For example, if a portion of a reference frame identified by a motion vector is not identical to the associated portion of the frame being encoded, the encoding module **130** may determine/identify differences (e.g., prediction errors) between the portion of the reference frame referenced by the motion vectors and the portion of the frame being encoded. An encoding parameter may be based on a number of motion vectors used by the encoding module **130** to encode a frame of content. For example, the encoding module **130** may use motion vectors to correlate a location of a portion of a reference frame to a portion of a frame being encoded.

(44) Additionally, the encoding module **130** may perform one or more transformation algorithms on content (e.g., video content, audio content, multimedia content, etc.). For example, the encoding module **130** may perform a discrete cosine transform (or any other transformation) on the content and/or a portion of a plurality of portions of the content. A transformation algorithm may comprise

expressing the content as a summation of functions (e.g., a summation of cosine functions). The functions may be related according to a formula. For example, each function may be raised to exponents, multiplied by coefficients, and/or provided arguments based on a summation formula. At least a portion of the plurality of portions of the content may be transformed according to a transformation algorithm. The coefficients of the functions resulting from the transformation algorithm may be encoded and provided as encoded content. For example, the encoding module **130** may encode only a portion of the coefficients resulting from the transformation algorithm.

(45) The encoding module **130** may quantize content (e.g., video content, audio content, multimedia content, etc.) and/or encoded data indicating the content (e.g., coefficients resulting from a transformation algorithm). Quantization may comprise converting content and/or encoded data into a smaller set of content and/or encoded data. For example, the coefficients may comprise an integer (e.g., 299792458) and/or a non-integer (e.g., 1.618033) real number. The encoding module **130** may quantize a coefficient by truncating, rounding, or otherwise reducing the number of digits in a number. For example, the example coefficient 1.618033 may be quantized to 1.618. The amount of quantization may be based on a quantization step size. A smaller quantization step size may result in the loss of less data than a larger quantization step size. For example, a larger quantization step size may result in a quantized coefficient of 1.61 and a smaller quantization step size may result in a quantized coefficient of 1.61803.

(46) An encoding parameter may comprise an encoding quantization level. The encoding quantization level may be based on the number of coefficients encoded from the transformation algorithm. For example, the encoding quantization level may be a measurement of the number of coefficients encoded for one or more frames (e.g., a plurality of images, a group of pictures, etc.) of encoded content. For example, the transformation algorithm may assign a coefficient to a number of values within range of a coefficient (e.g., coefficients of 1.1 and 0.9 could be represented as 1). Such ranges may be considered as quantization buckets or bins. The quantization level may be indicative of a number of pixel values assigned to a coefficient. The encoding parameter may comprise a number value indicating the quantization level (e.g. for a frame or group of frames). The encoding parameter may also comprise an average of the number values indicating the quantization level, an evaluation (e.g., high, medium, low) of the number value indicating a quantization level with respect to a threshold, and/or the like. For example, the quantization levels may be categorized as high, medium, or low based on the size of the ranges of coefficients (e.g., quantization buckets or bins) used in a transformation algorithm. The quantization levels may be categorized as high, medium, or low based on comparison of the ranges of the coefficients to average sizes of the ranges of coefficients. The quantization levels may be categorized as high, medium, or low based on the percentage of coefficients that reside in a dead zone. For example, a dead zone may be a range of values around a zero-output region of the quantization algorithm.

(47) An encoding parameter may comprise information based on one or more prediction errors determined by the encoding module **130** when encoding inter-coded frames, such as P-frames or B-frames. For example, a prediction error may comprise a measurement of differences between references frames and encoded frames. A prediction error may comprise a difference between pixel values (e.g., 0 through 255) of a reference frame as compared to a decoded frame. The encoding parameter may indicate a number of times the encoding module **130** determines/generates prediction errors when encoding one or more frames/portions (e.g., in a group of pictures) of content. For example, the encoding module **130** may determine/generate a prediction error when mapping one or more portions (e.g., pixels) of the reference frame of content to one or more portions of a frame (e.g. pixels) of the content being encoded. The encoding parameter may comprise a number of prediction errors (e.g., for one or more pixels, groups of pixels, frames, groups of frames, etc.), an average of the number of prediction errors, an evaluation (e.g., low, medium, high, etc.) of the number of prediction errors with respect to a threshold, and the like.

(48) A prediction error may be normalized against each candidate frame from a group of frames of

content to be encoded. The encoding module **130** may determine low, medium, and high values based on a comparison to standard deviations from a norm (e.g., mean or average value). For example, a prediction error may be considered low if the prediction error is below one standard deviation from the norm. A prediction error may be considered medium if the prediction error is within one standard deviation from the norm. A prediction error may be considered high if the prediction error is above one standard deviation from the norm.

(49) An encoding parameter may comprise a quantizing step size. The quantization step size may comprise a measure of the amount of information stored in each coefficient generated by a transformation operation. For example, the measurement may indicate a number of digits of the coefficient retained after the coefficients are truncated, rounded, or otherwise reduced. For example, a determined coefficient 1.618033 may be reduced in data size. A reduction to 1.6 may be indicated by a larger quantization step size than a reduction to 1.618. An encoding parameter may comprise one or more quantization step sizes determined by the encoding module **130**, an average of quantization step sizes determined by the encoding module **130**, an evaluation (e.g., low, medium, high) of values determined by the encoding module **130** with respect to a threshold, and/or the like. For example, the encoding module may determine a quantization step size as high, medium, or low based on comparison to an average quantization step size.

(50) An encoding parameter may comprise a comparison of an inter-coded frame to an intra-coded frame. For example, the encoding module **130** may compare the size (e.g., data storage size) of the intra-coded frame to the size of an inter-coded frame. An I-frame may be compared to one or more P-frames and/or B-frames. The size of the I-frame may be compared to the size of one or more P-frames and/or B-frames. For example, the size may be based on a number of bytes of the frame. An encoding parameter may comprise and/or be based on a comparison (e.g., the size of the difference) of intra-coded frame to more than one inter-coded frames. For example, an encoding parameter may be an average of comparisons for a group of frames of a plurality of frames (e.g., a plurality of portions of content, etc.). The encoding parameter may also comprise a value determined by the encoding module **130** by evaluating the comparison with respect to a threshold. For example, the encoding module **130** may determine the comparison of the intra-coded frame to the inter-coded frame to be low, medium, high, or the like. For example, B or P frames that are 0.4 the size of an I-frame or less may be considered low, values between 0.4 and 0.6 may be medium, and values above 0.6 may be considered high. The encoding module **130** may determine the ratios indirectly. For example, the number of prediction errors generated by the encoding module **130** in predicting/determining the P and B frames with motion vectors may be indicative of a higher relative size of the P and B frames compared to the I-frame.

(51) The content management device **128** (e.g., the encoding module **130**) may encode content and/or portions of a plurality of portions of the content (e.g., video content, audio content, multimedia content, etc.) by any method. An encoding parameter may indicate a level of complexity (e.g., low complexity, medium complexity, high complexity, etc.) to encode a portion of a plurality of portions of content (e.g., video content, audio content, multimedia content, etc.). The content management device **128** may comprise a complexity determination module **131**. The complexity determination module **131** may determine a level of complexity required to encode content. An encoding parameter may comprise/indicate a level of complexity to encode a portion of a plurality of portions of content that is determined by the complexity determination module **131**.

(52) The complexity determination module **131** may determine a level of complexity required for the encoding module **130** to encode content (e.g., video content, audio content, multimedia content, etc.) and/or a portion (e.g., a frame) of a plurality of portions (e.g., frames) of the content. For example, the complexity determination module **131** may determine one or more encoding parameters indicative of a level of complexity to encode content. An encoding parameter may comprise a complexity value related to an operation performed (e.g., performed by the encoding module **130**, etc.) in encoding the content. The complexity value may be a measurement of a

number of aspects (e.g., calculations, values, groupings, averages) of an operation performed by the encoding module **130** and/or results of the operation. For example, the level of complexity may be based on one or more of a prediction frame error associated with a portion (e.g., a frame) of content, a relative size of an inter-coded frame of the portion of the content with respect to an intra-coded frame of the portion of the content, or a number of motion vectors used by the encoding module **130** to encode a frame of the content. The complexity value may indicate a relative complexity (e.g., low complexity, medium complexity, high complexity, etc.) to the operation performed on other content.

(53) The complexity determination module **131** may determine a level of complexity required for the encoding module **130** to encode content (e.g., video content, audio content, multimedia content, etc.) and/or a portion (e.g., a frame) of a plurality of portions (e.g., frames) of the content by determining (e.g., counting) a number of motion vectors used by the encoding module **130** to encode a frame (and/or a group of frames) of a plurality of frames of content. The complexity determination module **131** may measure a size of one or more motion vectors of a frame (e.g., a portion) of content encoded by the encoding module **130**. For example, the complexity determination module **131** may calculate a distance between a portion of a reference frame and a portion of an encoded frame associated by a motion vector. The complexity determination module **131** may determine the distance by performing one or more calculations, such as: an absolute value operation, a magnitude operation, determining subcomponents of the motion vector, calculating a motion vector length based on Pythagorean's theorem, calculating the value of a trigonometric function (e.g., sine, cosine, tangent, cotangent), determining the difference between a coordinate at a point in the reference frame from a coordinate at a point in the encoded frame, and the like. The encoding parameter may comprise one or more determined values (e.g., number of motion vectors, length of motion vectors), an average of determined values, an evaluation (e.g., low, medium, high) of determined values with respect to a threshold, and/or the like. For example, motion vector values may be determined as high, medium, based on a comparison to an average (e.g., real-time average) distance of movement indicated by the motion vectors.

(54) The complexity determination module **131** may determine a level of complexity required for the encoding module **130** to encode content (e.g., video content, audio content, multimedia content, etc.) and/or a portion (e.g., a frame) of a plurality of portions (e.g., frames) of the content by any method. For example, the complexity determination module **131** may determine a level of complexity required for the encoding module **130** to encode content and/or a portion (e.g., a frame) of a plurality of portions (e.g., frames) of the content by determining by an amount of change between a plurality of pixels associated with a previous portion of the content and a corresponding plurality of pixels associated with a portion of the content. For example, a level of complexity to encode (e.g., for the encoding module **130** to encode) a portion of content may be a low level of complexity if an amount of change in pixels between the portions of the content does not satisfy a threshold, or the level of complexity to encode (e.g., for the encoding module **130** to encode) the portion of content may be a high level of complexity if the amount of change in pixels between the portions of content satisfies or exceeds the threshold.

(55) As previously described, content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). The content management device **128** may tag and/or associate each portion of the plurality of portions of the content with a content element of a plurality of content elements. A content element may be associated with attributes of content. The content management device **128** may comprise a content analysis module **132**. The content analysis module **132** may determine one or more attributes of content. For example, the content analysis module **132** may determine an attribute for each portion of a plurality of portions of content. An attribute of a portion of content may be and/or be associated with (e.g., indicate) a scene transition (e.g., scene change) between the portion of the content (e.g., a frame of the content) and another portion of the content (e.g., another frame of the

content).

(56) The content analysis module **132** may determine a scene transition (e.g., a scene change) between a portion of content (e.g., a frame of content) and another portion of the content (e.g., another frame of the content). The content analysis module **132** may determine a luminance value for a plurality of pixels associated with a previous portion (or post portion) of the content and a luminance value for a corresponding plurality of pixels associated with a portion of the content. For example, the content analysis module **132** may determine a plurality of luminance values associated with a plurality of pixels of a previous frame (or post frame) of content by determining a luminance value for each pixel of the plurality of pixels. The content analysis module **132** may determine a total luminance value associated with the previous frame (or post frame) of the content based on the plurality of luminance values associated with the plurality of pixels of the previous frame (or post frame) of the content. For example, the total luminance value may be a sum, average, or any other collective value of the plurality of luminance values associated with the plurality of pixels of the previous frame (or post frame) of the content. The content analysis module **132** may determine a total luminance value associated with a frame (e.g., a current frame) of the content by the same method described for determining the total luminance value associated with the previous (or post) frame.

(57) The content analysis module **132** may determine a scene transition (e.g., a scene change) between a portion of content (e.g., a frame of content) and another portion of the content (e.g., another frame of the content) by comparing a total luminance value associated with the portion of the content to a total luminance value associated with the other portion of the content (e.g., other frame of the content, previous frame of the content, post frame of the content, etc.). A difference in total luminance value may indicate a change from a full color scene of the content, to a black scene of the content. A scene transition associated with the content (e.g., portion of the content) may be indicated by the presence of a black screen. The content analysis module **132** may determine that a scene associated with the content is not transitioning if the total luminance value associated with the portion of the content is the same value, or if a difference between the total luminance values does not satisfy a threshold (e.g., the threshold may not be satisfied if the difference between the total luminance values is less than a value, such as of 2). The content analysis module **132** may determine that a scene associated with the content is transitioning if the total luminance value associated with the portion of the content is the different, or if a difference between the total luminance values satisfies or exceeds the threshold (e.g., the threshold may be satisfied if the difference between the total luminance values is equal to or greater than a value, such as a value of 2).

(58) The content analysis module **132** may use any method to determine a scene transition (e.g., a scene change) between a portion of content (e.g., a frame of content) and another portion of the content (e.g., another frame of the content). A scene transition may be indicated by a portion of content comprising and/or being associated with dialog and a previous or post portion of the content not comprising and/or being associated with dialog. The content analysis module **132** may determine whether a portion of content comprises and/or is associated with dialog by determining/analyzing closed captions (e.g., closed captioning data, metadata, etc.) associated with the content. For example, the content analysis module **132** may determine a scene transition associated with content based on closed captions (e.g., closed captioning data, metadata, etc.) associated with a previous (or post) portion of the content and a closed captions associated with a portion (e.g., current portion) of the content. For example, the content analysis module **132** may analyze attributes (e.g., metadata, associated files, etc.) of each portion of a plurality of portions of content and determine whether a respective portion of the content comprises and/or is associated with closed captions. The content analysis module **132** may determine a scene transition (e.g., a scene change) between portions of the content by determining whether consecutive portions of the content comprise and/or are associated with closed captions. For example, a scene transition may

be indicated by a previous (or post) portion of content comprising and/or being associated with closed captions (e.g., closed captioning data, metadata, etc.) and a portion of the content (e.g., a current portion of the content) not comprising and/or not being associated with closed captions.

(59) The content analysis module **132** may use any method to determine a scene transition (e.g., a scene change) between a portion of content (e.g., a frame of content) and another portion of the content (e.g., another frame of the content). A scene transition may be indicated by a portion of content comprising and/or being associated with an audio level (e.g., sound level) and a previous or post portion of the content comprising and/or being associated with a different audio level (e.g., sound level). The difference between audio levels may be based on a threshold. The content analysis module **132** may determine that a scene transition does not occur between portions of content if a difference between respective audio levels associated with the portions of the content does not satisfy a threshold. The content analysis module **132** may determine that a scene transition occurs between portions of content if a difference between respective audio levels associated with the portions of the content satisfy or exceed the threshold.

(60) The content analysis module **132** may determine additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with content. The content analysis module **132** may determine additional content related items associated with a portion of a plurality of portions of content (e.g., video content, audio content, multimedia content, etc.). For example, the content analysis module **132** may determine that a portion of content is associated with an advertisement (e.g., a portion of an advertisement, a beginning of an advertisement, an end of an advertisement, etc.). The content analysis module **132** may determine that the portion of the content is associated with the advertisement (e.g., a portion of the advertisement, a beginning of the advertisement, an end of the advertisement, etc.) based on metadata or any other indicator (e.g., a watermark, a resource location, etc.) of the advertisement.

(61) As previously mentioned, the content management device **128** may tag and/or associate each portion of a plurality of portions of content (e.g., video content, audio content, multimedia content, etc.) with a content element of a plurality of content elements. As described, a content element may be associated with and/or indicate one or more of an encoding parameter (e.g., a level of complexity to encode, etc.) associated with the respective portion of the content, attributes of content (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content, additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with the respective portion of the content, or the like. The content management device **128** may use a content element determine a bitrate to associate with a portion of content.

(62) The content management device **128** may package content (e.g., video content, audio content, multimedia content, etc.) for delivery to a network device (e.g., a content player, a media device, a smart device, a mobile device, a computing device, etc.). The content management device **128** may associate content with a content element to enable quality consumption of the content. For example, quality (e.g., picture quality, sound quality, visual quality, etc.) consumption of content may be content that may be consumed (e.g., displayed, watched, listened to, etc.) without perceived changes in video quality (e.g., no image artifacts, no blurred images, no unintentionally paused and/or delayed video, etc.), audio quality (e.g., no unintentional changes in volume, no changes in sound clarity, etc.), and or the like. The content management device **128** may use a plurality of content elements to determine a plurality of bitrates to associate with a plurality of portions of content such that each portion of the plurality of portions of the content is associated with a content element and one or more bitrates, such as a low bitrate, a medium bitrate, a high bitrate, or the like. The content management device **128** may determine that any bitrate is a low bitrate, a medium bitrate, or a high bitrate. The content management device **128** may determine that any bitrate is a low bitrate, a medium bitrate, or a high bitrate by any method, such as based on network (e.g., the network **116**, etc.) conditions (e.g., latency in a portion/channel of the network, network

congestion, provisioned information and/or data transfer rates, etc.), user determined bitrate classifications (e.g., user defined bitrate ranges from high to low, etc.), dynamically determined bitrate classifications, or the like. The content management device **128** may associate any bitrate and/or plurality of bitrates to any portion of content based on a content element associated with the portion of the content.

(63) The content management device **128** may associate a bitrate and/or a plurality of bitrates to a portion of a plurality of portions of content based on an associated level of complexity to encode the portion of the content (as indicated by an associated content element). For example, an impact on a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of a portion of content may be caused by network congestion (or any other cause) and may result in image artifacts, blurred images, illegible sound, paused/frozen content, and the like associated with a portion of content. A low level of complexity to encode a portion of content may indicate that the portion of the content may be provided to a network device (e.g., a content player, a media device, a smart device, a mobile device, a computing device, etc.) at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content by the network device (a user of the network device) will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. The content management device **128** may determine that an encoding parameter associated/indicated by a content element indicates that a portion of content is associated with a level of complexity (e.g., a low level of complexity, a medium level of complexity, a high level of complexity, etc.) to encode (e.g., encoded by the encoding module **130** and/or any other device) the portion of the content. The content management device **128** may associate a low bitrate and/or a plurality of low bitrates to a portion of content associated with a content element (e.g., encoding parameter) that indicates a low level of complexity to encode the portion of the content. A high level of complexity to encode a portion of content may indicate that the portion of the content should be provided to a network device at a high bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content by the network device (a user of the network device) will be impacted if the portion of the content is not provided and/or consumed at a high bitrate. The content management device **128** may associate a high bitrate and/or a plurality of high bitrates to a portion of content associated with a content element (e.g., encoding parameter) that indicates a high level of complexity to encode the portion of the content. The content management device **128** may associate any bitrate and/or plurality of bitrates to any portion of content based on a content element associated with the portion of the content.

(64) The content management device **128** may associate a bitrate and/or a plurality of bitrates to each portion of a plurality of portions of content based on attributes (e.g., a scene transition, a scene change, etc.) of the content associated with the respective portion of the content and indicated by a content element. For example, the content management device **128** may associate a low bitrate and/or a plurality of low bitrates to a portion of content associated with a content element that indicates a scene transition (e.g. scene change, etc.) associated with the portion of the content. A quality of consumption associated with the portion of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate because a scene transition may be associated with a visually black screen, a lack of dialog/audio, or the like occurring at the point (e.g., portion of content) of the scene transition. The content management device **128** may associate a medium/high bitrate and/or a plurality of medium/high bitrates with portions of content associated with a content element that indicates a scene transition (e.g. scene change, etc.) associated with the portions of the content does not occur. A scene transition that does not occur during a portion of content may be an indication (e.g., indicated by an associated content element, etc.) that the portion of content is associated with an important scene of the content, such as a scene/portion of the content where a quality of consumption associated with the scene/portion will be perceived as impacted if the scene/portion of

the content is consumed at a low bitrate. The content management device **128** may associate any bitrate and/or plurality of bitrates to any portion of content based on a content element associated with the portion of the content.

(65) The content management device **128** may associate a bitrate and/or a plurality of bitrates to each portion of a plurality of portions of content based on additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with a respective portion of the content and indicated by a content element. For example, the content management device **128** may associate a bitrate (e.g., a low bitrate, a medium bitrate, a high bitrate, etc.) and/or a plurality of bitrates to a portion of content associated with a content element that indicates that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portion of the content. The advertisement may be associated with the portion of the content based on metadata associated with the advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) and/or the portion of the content. The content management device **128** may associate any bitrate and/or any plurality of bitrates with portions of content associated with a content element that indicates that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portion of the content.

(66) The content management device **128** may store content and/or provide the content to storage (e.g., a database, a cloud-based storage, etc.). The content management device **128** may store the content and/or provide the content to storage such that each portion of the content is stored with a content element and at a bitrate and/or a plurality of bitrates determined based on the content element.

(67) The content management device **128** may provide content to a network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.). The content management device **128** may provide the content to the network device based on a request for the content. The content management device **128** may provide the content to the network device such that each portion of the content is provided with a content element and/or at a bitrate and/or a plurality of bitrates determined based on the content element. The content may be provided to the network device and/or any other device as streaming content, downloaded content, referenced and retrieved content, or the like. The content management device **128** may provide the content in a form comprising, an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like. For example, the content management device **128** may provide the content to the network device and/or any other device as a manifest file comprising a plurality of references to the content. The content management device **128** may provide the content to the network device and/or any other device as a manifest file comprising a plurality of content elements and a plurality of references to the content such that each portion of the content is associated with a bitrate and/or a plurality of bitrates determined based on a content element of the plurality of content elements. The content management device **128** may provide the content to the network device and/or any other device by any method. The network device and/or any other device may retrieve/receive the content by any method.

(68) In an aspect, the content management device **128** may associate a bitrate and/or a plurality of bitrates to a portion of a plurality of portions of content based on consumption of the content (e.g., access/retrieval of the content, display of the content, etc.). For example, the content management device **128** may associate a bitrate and/or a plurality of bitrates to a portion of a plurality of portions of content based on an amount of network devices (e.g., content players, the media device **120**, smart devices, the mobile device **124**, computing devices, the display device **121**, etc.) accessing a portion of a plurality of portions of content, an amount of network devices displaying a portion of a plurality of portions of content, and/or an amount of network devices requesting a portion of a plurality of portions of content. The content management device **128** may associate a low bitrate and/or a plurality of low bitrates to a portion of content that is associated with

consumption that does not satisfy a threshold. The content management device **128** may associate a high bitrate and/or a plurality of high bitrates to a portion of content that is associated with consumption that satisfies and/or exceeds a threshold. The content management device **128** may associate any bitrate and/or plurality of bitrates to any portion of content based on any level (e.g., threshold level, etc.) of consumption of the respective portion of the content.

(69) In an aspect, the content management device **128** can, based on consumption of content (e.g., access/retrieval of the content, display of the content, etc.), use a content element associated with a portion of a plurality of portions of the content to associate a bitrate and/or a plurality of bitrates to the portion of the plurality of portions of content. For example, the content management device **128** may determine consumption of a portion of the plurality of portions of content. The content management device **128** may determine that a content element associated with the portion of the content and/or a previous/post portion of the content indicates a scene transition (e.g. scene change, etc.) and associate a high bitrate (or a low bitrate) and/or a plurality of high bitrates (or a plurality of low bitrates) with the portion of the content and/or a previous/post portion of the content so bitrate change is not perceived (e.g., content provided in a best possible quality, etc.) by a user(s) of a network device(s) associated with the consumption. In an aspect, the content management device **128** may associate a bitrate and/or a plurality of bitrates to a portion of a plurality of portions of content based on attributes associated with a content delivery network (e.g., the network **116**, etc.), such as network latency/congestion, bandwidth parameters/availability, and/or the like. For example, the content management device **128** may associate a bitrate and/or a plurality of bitrates to a portion of a plurality of portions of content based on an amount of bandwidth available in a network, determined latency/congestion in the network, and/or network parameters (e.g., a committed information rate, etc.) when a portion of a plurality of portions of content is accessed, requested, retrieved, and/or the like by a network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) and/or any other device. In an aspect, the content management device **128** may associate a bitrate and/or a plurality of bitrates to each portion of a plurality of portions of content based on any factor and/or determination by the content management device **128**, a network device, and/or any other device.

(70) A network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) and/or any other device may consume content and/or portions of the content in a best possible quality (e.g., based on a best possible bitrate, etc.) according to a plurality of content elements associated with the content (e.g., a plurality of portions of the content). The best possible quality may be a quality (e.g., picture quality, sound quality, visual quality, etc.) of the content with image artifacts, blurred images, illegible sound, paused/frozen content, or the like reduced and/or negligible. To ensure that each portion of the plurality of portions of the content is consumed at the best quality, the network device may determine/select a bitrate of a plurality of bitrates based on respective content elements of the plurality of content elements associated with each portion of the plurality of portions of the content. The network device may use a content element associated with a portion of content to determine/select a bitrate of the plurality of bitrates associated with the content that will enable the content to be consumed at the best quality. For example, if a network device (or a user associated with the network device) is consuming content, the network device may automatically select/adjust a bitrate associated with a portion of the content because a content element indicates that the quality of consumption will not be impacted by a selection/adjustment of a bitrate, such as a selection/adjustment to a lower bitrate, or a higher bitrate compared to a bitrate at which other portions of the content are/can be consumed.

(71) FIG. 2 is a diagram of variable bitrate content. The content **200** may be video content, audio content, multimedia content, or any other type of content. The content **200** may comprise portions **201, 202, 203, 204, 205, 206, 207**. The portions **201, 202, 203, 204, 205, 206, 207** may be frames, segments, fragments, or the like of the content **200**. Each of the portions **201, 202, 203, 204, 205,**

206, 207 may be based on and/or associated with an interval (e.g., 2-second interval, etc.) of the content **200**.

(72) The content **200** may comprise a plurality of content elements **208, 209, 210, 211, 212, 213, 214**. Each of the content elements **208, 209, 210, 211, 212, 213, 214** may be associated with and/or indicate an encoding parameter associated with the respective portions **201, 202, 203, 204, 205, 206, 207** of the content **200**. For example, the content element **208** may be associated with and/or indicate an encoding parameter associated with the portion **201** of the content **200**, the content element **209** may be associated with and/or indicate an encoding parameter associated with the portion **202** of the content **200**, and so on. An encoding parameter associated with a content portion may be based on and/or indicate a level of complexity to encode the content portion. A level of complexity to encode a content portion may be indicated, for example, as a low complexity, a medium complexity, a high complexity, and/or the like.

(73) A level of complexity associated with each content portion **201, 202, 203, 204, 205, 206, 207** may be based on a prediction frame error associated with the respective content portion during encoding and/or compression of the content **200**. For example, the content **200** may comprise intra-coded frames (I-frames). I-frames can, during encoding/compression, be processed independent from frames that precede and/or come after them. Any of the content portions **201, 202, 203, 204, 205, 206, 207** may be an intra-coded frame (I-frame). An I-frame may contain/store all information/data necessary to display the frame (e.g., the content portions **201, 202, 203, 204, 205, 206, 207**, etc.). The content **200** may comprise predictive frames (P-frames). Any of the content portions **201, 202, 203, 204, 205, 206, 207** may be a P-frame. P-frames may follow I-frames and be encoded with and/or comprise information/data of any content changes from the preceding I-frame (e.g., a change in color, a change in content, etc.). A prediction frame error may be generated based on a predicted (e.g., expected, etc.) measurement of difference between a reference frame (e.g., I-frame, etc.) and an encoded frame (e.g., P-frame, etc.) failing to coincide with an actual measurement of difference between the reference frame (e.g., I-frame, etc.) and the encoded frame (e.g., P-frame, etc.). A prediction error may be based on a difference between pixel values (e.g., 0 through 255) of an I-frame and pixel values of an encoded frame (P-frame, etc.). For example, if content portion **201** is an I-frame and content portion **202** is a P-frame, and during encoding of the content portion **202**, an amount of change in color (e.g., pixel color, etc. . . .) does not coincide with a predicted amount of change in color, then a prediction frame error may be generated. An encoding parameter associated with a content portion (e.g., the content portions **201, 202, 203, 204, 205, 206, 207**, etc.) may comprise a number of prediction errors (e.g., for one or more pixels, groups of pixels, frames, groups of frames, etc.), an average of the number of prediction errors, an evaluation (e.g., low, medium, high, etc.) of the number of prediction errors with respect to a threshold, and the like. A level of complexity associated with a content portion (e.g., a frame of content, the content portions **201, 202, 203, 204, 205, 206, 207**, etc.) may be based on an amount of prediction frame errors associated with the frame. For example, a content portion with an amount of prediction frame errors below a threshold amount of prediction frame errors may be associated with a low level of complexity, a content portion with an amount of prediction frame errors equaling the threshold amount of prediction frame errors may be associated with a medium level of complexity, and a content portion with an amount of prediction frame errors exceeding the threshold amount of prediction frame errors may be associated with a high level of complexity.

(74) A level of complexity associated with each content portion **201, 202, 203, 204, 205, 206, 207** may be based on a number of motion vectors associated with a frame of the respective content portion. For example, each content portion **201, 202, 203, 204, 205, 206, 207** may comprise inter-coded frames and intra-coded frames. An inter-coded frame may be divided into blocks known as macroblocks (e.g., MCU blocks, H.261, MPEG-1 Part 2, H.262/MPEG-2 Part 2, H.263, MPEG-4 Part 2, and H.264/MPEG-4 AVC, etc.). Macroblocks may be processing units used during encoding/compression according to linear block transforms, such as a discrete cosine transform

(DCT), for example. A macroblock may consist of 16×16 samples, may be subdivided into transform blocks, and may be further subdivided into prediction blocks. After that, instead of directly encoding the raw pixel values for each block, the encoder will try to find a block similar to the one it is encoding on a previously encoded frame, referred to as a reference frame. This process is done by a block matching algorithm. If the encoder succeeds on its search, the block could be encoded by a vector, known as motion vector, which points to the position of the matching block at the reference frame. The process of motion vector determination is called motion estimation.

(75) The level of complexity may be based on and/or determined by an amount of change between a plurality of pixels associated with a previous portion of the content **200** and a corresponding plurality of pixels associated with the respective portion of the content **200**. For example, a level of complexity to encode a portion of content **200** may be low if an amount of change in pixels between the portions of content **200** does not satisfy a threshold, or the level of complexity to encode the portion of content may be high if the amount of change in pixels between the portions of content satisfies or exceeds the threshold. An amount of change in pixels between the portions of content **200** that does not satisfy a threshold may indicate that the respective portion of the content is not complex, is associated with a scene-transition and/or black screen, is associated with one or more advertisements, and/or the like. An amount of change in pixels between the portions of content **200** that satisfy a threshold may indicate that the respective portion of the content is complex, contains detailed images, contains enhanced audio attributes, is associated with a quantity of items/objects that may be displayed, and/or any other attribute of the content.

(76) A level of complexity associated with a content portion may be based on a relative size of an inter-coded frame with respect to an intra-coded frame. An inter-coded frame may be a frame of content that has been encoded/compressed based on data (e.g., metadata, pixel orientation, pixel luminance, etc.) from many frames immediately preceding or succeeding the respective frame. An intra-coded frame may be a frame of content that has been encoded/compressed without regard/reference to other frames of the content. Any of the content portions **201, 202, 203, 204, 205, 206, 207** may be an inter-coded frame. Any of the content portions **201, 202, 203, 204, 205, 206, 207** may be an intra-coded frame. For example, the content portion **202** may be an inter-coded frame, and the content portion **203** may be an intra-coded frame. A level of complexity associated with the content portion **202** may be based on a relative size of the content portion **202** with respect to the content portion **203**. Any of the content portions **201, 202, 203, 204, 205, 206, 207** may be an inter-coded frame. Any of the content portions **201, 202, 203, 204, 205, 206, 207** may be an intra-coded frame.

(77) Each content element **208, 209, 210, 211, 212, 213, 214** may be used to determine a bitrate (e.g., a low bitrate, a medium bitrate, a high bitrate, etc.) or a plurality of bitrates (e.g., a range of low bitrates, a range of medium bitrates, a range of high bitrates, etc.) to associate each of the portions **201, 202, 203, 204, 205, 206, 207** of the content **200**.

(78) The content elements **208, 209, 210, 211, 212, 213, 214** may be associated with attributes of content **200** (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content. For example, the content element **211** may be and/or be associated with metadata that indicates and/or causes a scene transition between the portion **203** of content **200** and the portion **204** of the content **200**. The scene transition may be associated with and/or identified by a visually black screen (as indicated by **225** and **226**), a lack of dialog/audio, or the like occurring at the portion **203** of content **200** and the portion **204** of the content **200** and at the portion **206** of content **200** and the portion **207** of the content **200**. As another example, the content elements **208, 209, 210, 211, 212, 213, 214** may be and/or be associated with metadata that associates the respective portions **201, 202, 203, 204, 205, 206, 207** of the content **200** with one or more advertisements. As another example, the content elements **208, 209, 210, 211, 212, 213, 214** may be and/or be associated with metadata that associates an audio level with the respective portions **201, 202, 203, 204, 205, 206, 207**.

(79) The content elements **208, 209, 210, 211, 212, 213, 214** may be used to determine (e.g., vary, adapt, etc.) a bitrate for and/or to associate with the respective portions **201, 202, 203, 204, 205, 206, 207** of the content **200**. Each content portion **201, 202, 203, 204, 205, 206, 207** may be associated with a bitrate, indicated by **215, 216, 217, 218, 219, 220, 221**. The bitrates **215, 216, 217, 218, 219, 220, 221** may be associated with the respective portions **201, 202, 203, 204, 205, 206, 207** of the content **200** based on attributes (e.g., a scene transition **225**, a scene transition **226**, etc.) of the content **200** associated with the respective portions **201, 202, 203, 204, 205, 206, 207** and indicated by a content elements **208, 209, 210, 211, 212, 213, 214**. For example, the portions **217** and **220** may be associated with a low bitrate and/or a plurality of low bitrates (e.g., a range of low bitrates indicted by **222**) because the content element **210** and **213**, respectively, indicates a scene transition (e.g. scene change, etc.) associated with the portions **203** and **206** of the content **200**. A quality of consumption associated with the portions **203** and **206** of the content **200** may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portions **203** and **206**) by a low bitrate because a scene transition (e.g., **225** and **226**) may be associated with a visually black screen, a lack of dialog/audio, or the like occurring at the portions **203** and **206**. A medium bitrate (e.g., a range of medium bitrates indicted by **223**) may be associated with the portions **201, 202, 207** because the respective content elements **208, 209, 214** indicate that a scene transition (e.g. scene change, etc.) associated with the portions **201, 202, 207** does not occur. Portions **204** and **205** of the content **200** may be associated with a high bitrate (e.g., a range of high bitrates indicted by **224**) because the portions **204** and **205** are after the scene transition **225**. A portion of content occurring after a scene transition may be an indication (e.g., indicated by an associated content element, etc.) that the portions **204** and **205** of the content **200** are associated with an important scene of the content **200**, such as a scene/portion of the content **200** where a quality of consumption associated with the scene/portion will be perceived as impacted if the scene/portion of the content is consumed at a low bitrate or a medium bitrate. Any bitrate and/or plurality of bitrates may be associated with any portion of the content **200** based on a content element associated with the portion of the content **200**.

(80) A bitrate or a range of bitrates (e.g., the low bitrates **222**, the medium bitrates **223**, the high bitrates **224**, etc.) may be associated with the portions **201, 202, 203, 204, 205, 206, 207** of the content **200** based on additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with the portions **201, 202, 203, 204, 205, 206, 207** of the content **200** indicated by the content elements **208, 209, 210, 211, 212, 213, 214**. For example, a bitrate (e.g., a low bitrate, a medium bitrate, a high bitrate, etc.) and/or a plurality of bitrates (e.g., the low bitrates **222**, the medium bitrates **223**, the high bitrates **224**, etc.) may be associated with portions **201, 202, 203, 204, 205, 206, 207** of the content **200** if the content elements **208, 209, 210, 211, 212, 213, 214** indicate that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portions **201, 202, 203, 204, 205, 206, 207**. The advertisement may be associated with the portions **201, 202, 203, 204, 205, 206, 207** based on metadata associated with the advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) and/or the portions **201, 202, 203, 204, 205, 206, 207** of the content **200**. Any bitrate and/or any plurality of bitrates may be associated with the portions **201, 202, 203, 204, 205, 206, 207** of the content **200** if a content element indicates that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portions **201, 202, 203, 204, 205, 206, 207**.

(81) FIG. 3 is a flowchart of an example method **300** for providing variable bitrate content. Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). At **310** a level of encoding complexity may be determined for each portion of a plurality of portions of content. A device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may determine the

level of encoding complexity for each portion of a plurality of portions of content.

(82) The plurality of portions of the content may be based on a determined amount of time (e.g., time window, time duration, etc.). For example, each portion of the plurality of portions of the content may be based on and/or associated with an interval (e.g., 2-second interval, etc.) of the content. The interval may comprise and/or be associated with a frame (e.g., a portion) of the content, a group of frames of the content, a block of the content, or the like.

(83) An encoding parameter may be determined with each portion of the plurality of portions of the content. The encoding parameter can indicate a level of complexity (e.g., low complexity, medium complexity, high complexity, etc.) to encode (e.g., for the content management device **128** and/or any other device to encode) the respective portion of the content. The level of complexity can be based on one or more of a prediction frame error associated with the respective portion of the content, a relative size of an inter-coded frame of the respective portion of the content with respect to an intra-coded frame of the respective portion of the content, or a number of motion vectors to encode a frame of the respective portion of the content. The level of complexity can be based on a numerical count, an average, a categorization, or the like, of information about encoding a portion of the content. The level of complexity can be based on and/or determined by an amount of change between a plurality of pixels associated with a previous portion of the content and a corresponding plurality of pixels associated with the respective portion of the content. For example, a level of complexity to encode a portion of content can be low if an amount of change in pixels between the portions of content does not satisfy a threshold, or the level of complexity to encode the portion of content can be high if the amount of change in pixels between the portions of content satisfies or exceeds the threshold.

(84) The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send the content to the user device such that each portion of the content is associated with a level of encoding complexity and/or at a bitrate and/or a plurality of bitrates determined based on level of encoding complexity. The content may be provided to the network device and/or any other device as streaming content, downloaded content, referenced and retrieved content, or the like. The device may provide the content in a form comprising, an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like. For example, the device may provide the content to the user device and/or any other device as a manifest file comprising a plurality of references to the content. The device may provide the content to the user device and/or any other device as a manifest file comprising a plurality of references to the content such that each portion of the content is associated with a bitrate and/or a plurality of bitrates determined based on a level of encoding complexity. The device may provide the content to the user device and/or any other device by any method. The user device and/or any other device may retrieve/receive the content by any method.

(85) At **320** one or more portions of the plurality of portions of the content that have a level of encoding complexity that satisfies a threshold may be sent at a bitrate. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send, at a first bitrate, one or more portions of the plurality of portions of the content that have a level of encoding complexity that satisfies a threshold to a user device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.).

(86) The level of encoding complexity for a portion of the content may satisfy a threshold when the portion of the content is associated with a complex scene, contains many detailed images, is associated with enhanced audio, and/or any other attribute of the portion of the content. The first bitrate may be a high bitrate. A high/higher bitrate may improve a quality of display of the portion of the content. The user device may consume the content and/or portions of the content in a best possible quality based on the high bitrate. The best possible quality may be a quality (e.g., picture quality, sound quality, visual quality, etc.) of the content with image artifacts, blurred images, illegible sound, paused/frozen content, or the like reduced and/or negligible. To ensure that the each

portion of the plurality of portions of the content is consumed at the best quality, the user device may determine/select the first bitrate of a plurality of bitrates based on the level of encoding complexity associated with the portion of the content.

(87) At **330** one or more portions of the plurality of portions of the content that have a level of encoding complexity that does not satisfy a threshold may be sent at a bitrate. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send, at a second bitrate, one or more portions of the plurality of portions of the content that have a level of encoding complexity that does not satisfy the threshold to the user device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.).

(88) The level of encoding complexity for a portion of the content may not satisfy the threshold when the portion of the content is associated with a non-complex scene, contains basic and/or monochromatic images, is associated with a scene transition, and/or any other attribute of the portion of the content. The second bitrate may be a low bitrate. A level of encoding complexity for a portion of the content that does not satisfy the threshold may indicate that the portion of the content may be sent to the user device at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. The level of encoding complexity for a portion of the content may not satisfy the threshold when the portion of the content is associated with a non-complex scene, contains basic and/or monochromatic images, contains a lack of dialog/audio, and/or any other attribute of the portion of the content. A quality of consumption associated with the portion of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate.

(89) FIG. **4** is a flowchart of an example method **400** for providing variable bitrate content. Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). At **410** scene-level metadata may be determined for each portion of a plurality of portions of content. A device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may determine the scene-level metadata for each portion of a plurality of portions of content.

(90) The plurality of portions of the content may be based on a determined amount of time (e.g., time window, time duration, etc.). For example, each portion of the plurality of portions of the content may be based on and/or associated with an interval (e.g., 2-second interval, etc.) of the content. The interval may comprise and/or be associated with a frame (e.g., a portion) of the content, a group of frames of the content, a block of the content, or the like.

(91) Scene-level metadata may be determined with each portion of the plurality of portions of the content. The scene-level metadata may indicate whether a scene associated with a portion of the plurality of portions of the content is complex or non-complex. Whether a scene is complex or non-complex may be based on one or more advertisements associated with the portion of the content, audio attributes of the content, and/or any other attribute associated with the content and/or the portion of the content.

(92) The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send the content to the user device such that each portion of the content is associated with scene-level metadata and/or at a bitrate and/or a plurality of bitrates determined based on scene-level metadata. The content may be provided to the network device and/or any other device as streaming content, downloaded content, referenced and retrieved content, or the like. The device may provide the content in a form comprising, an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like. For example, the device may provide the content to the user device and/or any other device as a manifest file comprising a plurality of references to the content. The device may

provide the content to the user device and/or any other device as a manifest file comprising a plurality of references to the content such that each portion of the content is associated with a bitrate and/or a plurality of bitrates determined based on scene-level metadata associated with each portion of the content. The device may provide the content to the user device and/or any other device by any method. The user device and/or any other device may retrieve/receive the content by any method.

(93) At **420** one or more portions of the plurality of portions of the content that have scene-level metadata indicative of content that is complex may be sent at a bitrate. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send, at a first bitrate, one or more portions of the plurality of portions of the content that scene-level metadata indicative of content that is complex to a user device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.). Scene-level metadata indicative of content that is complex may be based on one or more of pixel illumination, an audio level, or closed captioning information associated with a portion of the content. The first bitrate may be a high bitrate. A high/higher bitrate may improve a quality of display of the portion of the content. The user device may consume the content and/or portions of the content in a best possible quality based on the high bitrate. The best possible quality may be a quality (e.g., picture quality, sound quality, visual quality, etc.) of the content with image artifacts, blurred images, illegible sound, paused/frozen content, or the like reduced and/or negligible. To ensure that the each portion of the plurality of portions of the content is consumed at the best quality, the user device may determine/select the first bitrate of a plurality of bitrates based on the scene-level metadata associated with the portion of the content.

(94) At **430** one or more portions of the plurality of portions of the content that have scene-level metadata indicative of content that is not complex may be sent at a bitrate. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send, at a second bitrate, one or more portions of the plurality of portions of the content that have scene-level metadata indicative of content that is not complex to the user device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.).

(95) The scene-level metadata may indicate that content that is not complex when the portion of the content is associated with a non-complex scene, contains basic and/or monochromatic images, is associated with a scene transition, and/or is associated with an advertisement. The scene-level metadata may indicate that content that is not complex based on pixel illumination (e.g., chromatic scenes, monochromatic scenes, etc.), an audio level, or closed captioning information associated with the portion of the content, and/or any other attribute of the portion of the content. The second bitrate may be a low bitrate. Scene-level metadata that is not complex may indicate that the portion of the content may be sent to the user device at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. A quality of consumption associated with the portion of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate.

(96) FIG. 5 is a flowchart of an example method **500** for providing variable bitrate content. Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). The plurality of portions of the content may be based on a determined amount of time (e.g., time window, time duration, etc.). For example, each portion of the plurality of portions of the content may be based on and/or associated with an interval (e.g., 2-second interval, etc.) of the content. The interval may comprise and/or be associated with a frame (e.g., a portion) of the content, a group of frames of the content, a block of the content, or the like. Each portion of the plurality of portions of the content may be and/or be associated with a packet of

the content. At **510** a first packet of content may be received. A device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may receive the first packet and the first packet may comprise a level of encoding complexity associated with the portion of the content. The device may receive the first packet from a content source (e.g., content source **127**) and/or any other source/device. The device may receive the first packet from a content source (e.g., content source **127**) and/or any other source/device based on a request for the content, such as a request from a user device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) for the content. Each packet of the content may comprise an encoding parameter. The encoding parameter can indicate the level of complexity (e.g., low complexity, medium complexity, high complexity, etc.) to encode (e.g., for the content management device **128** and/or any other device to encode) the respective portion/packet of the content. The level of complexity can be based on one or more of a prediction frame error associated with the first packet of the content, a relative size of an inter-coded frame of the first packet of the content with respect to an intra-coded frame of the first packet of the content, or a number of motion vectors to encode the first packet of the content. The level of complexity can be based on a numerical count, an average, a categorization, or the like, of information about encoding the first packet of the content. The level of complexity can be based on and/or determined by an amount of change between a plurality of pixels associated with a previous packet of the content and a corresponding plurality of pixels associated with the first packet of the content. A level of complexity to encode the first packet of the content can be low if an amount of change in pixels between the first packet of the content and other packets of the content does not satisfy a threshold, or the level of complexity to encode the portion of content can be high if the amount of change in pixels between the first packet of the content and other packets of the content satisfies or exceeds the threshold.

(97) At **520**, a bitrate may be assigned to and/or determined for the first packet of the content. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may assign (determine) a first bitrate for the first packet of the content based on the level of encoding complexity of the first packet satisfying a threshold. The level of encoding complexity of the first packet of the content may satisfy the threshold when the of the first packet the content is associated with a complex scene, contains many detailed images, is associated with enhanced audio, and/or any other attribute of the portion of the content. The first bitrate may be a high bitrate. A high/higher bitrate may improve a quality of display of the first packet of the content. The user device may consume the content and/or first packet of the content in a best possible quality based on the first bitrate. The best possible quality may be a quality (e.g., picture quality, sound quality, visual quality, etc.) of the content with image artifacts, blurred images, illegible sound, paused/frozen content, or the like reduced and/or negligible.

(98) When the first packet (or any other packet) of the content is associated with a level of encoding complexity that does not satisfy the threshold, the first packet (or any other packet) may be assigned a second bitrate. The level of encoding complexity for the first packet (or any other packet) of the content may not satisfy the threshold when the first packet (or any other packet) of the content is associated with a non-complex scene, contains basic and/or monochromatic images, is associated with a scene transition, and/or any other attribute of the portion of the content. The second bitrate may be a low bitrate. A level of encoding complexity for the first packet (or any other packet) of the content that does not satisfy the threshold may indicate that the first packet (or any other packet) of the content may be sent to the user device at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. The level of encoding complexity for the first packet (or any other packet) of the content may not satisfy the threshold when the first packet (or any other packet) of the content is associated with a non-complex scene, contains basic and/or monochromatic images, contains a lack

of dialog/audio, and/or any other attribute of the portion of the content. A quality of consumption associated with the first packet (or any other packet) of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate.

(99) At **530** a second packet of content may be received. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may receive the second packet and the second packet may comprise scene-level metadata associated with the portion of the content. The device may receive the second packet from the content source (e.g., content source **127**) and/or any other source/device. Each packet of the content may comprise scene-level metadata. The scene-level metadata may indicate whether a scene associated with a portion of the plurality of portions of the content is complex or non-complex. Whether a scene is complex or non-complex may be based on one or more advertisements associated with the portion of the content, audio attributes of the content, and/or any other attribute associated with the content and/or the portion of the content.

(100) At **540**, a bitrate may be assigned to and/or determined for the second packet of the content. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may assign (determine) the first bitrate for the second packet of the content based on the scene-level metadata indicating that the packet (portion) of the content that is complex. Scene-level metadata indicative of content that is complex may be based on one or more of pixel illumination, an audio level, or closed captioning information associated with a portion of the content. As described, the first bitrate may be a high bitrate. A high/higher bitrate may improve a quality of display of the portion of the content. The user device may consume the content and/or portions of the content in a best possible quality based on the high bitrate. The best possible quality may be a quality (e.g., picture quality, sound quality, visual quality, etc.) of the content with image artifacts, blurred images, illegible sound, paused/frozen content, or the like reduced and/or negligible. To ensure that the packet of the content is consumed at the best quality, the user device may determine/select the first bitrate of a plurality of bitrates based on the scene-level metadata associated with the portion of the content.

(101) When the second packet (or any other packet) of the content is associated with scene-level metadata that is not complex, the second packet (or any other packet) may be assigned the second bitrate. The scene-level metadata may indicate that content that is not complex when the portion of the content is associated with a non-complex scene, contains basic and/or monochromatic images, is associated with a scene transition, and/or is associated with an advertisement. The scene-level metadata may indicate that content that is not complex based on pixel illumination (e.g., chromatic scenes, monochromatic scenes, etc.), an audio level, or closed captioning information associated with the portion of the content, and/or any other attribute of the portion of the content. As described, the second bitrate may be a low bitrate. Scene-level metadata that is not complex may indicate that the portion of the content may be sent to the user device at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. A quality of consumption associated with the portion of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate.

(102) At **540**, the first packet and the second packet may be sent at the first bitrate. The device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may send the first packet and the second packet of the content to the user device at the first bitrate. The first packet and the second packet of the content may be provided to the user device and/or any other device as streaming content, downloaded content, referenced and retrieved content, or the like. The device may provide the first packet and the second packet of the content in a form comprising, an index file, a multimedia file, a text file, code, a script, metadata, resource

location(s), combinations thereof, or the like. For example, the device may send/provide the first packet and the second packet of the content to the user device and/or any other device as a manifest file comprising a plurality of references to the first packet, the second packet, and/or any other packet of the content. The device may send/provide the first packet, the second packet, and/or any other packet of the content to the user device and/or any other device as a manifest file comprising a plurality of references to the first packet, the second packet, and/or any other packet of the content such that each portion of the content is associated with the first bitrate, the second bitrate, and/or any other bitrate. The device may provide the first packet, the second packet, and/or any other packet of the content to the user device and/or any other device by any method. The user device and/or any other device may retrieve/receive the first packet, the second packet, and/or any other packet of the content by any method.

(103) FIG. 6 is a flowchart of an example method **600** for providing variable bitrate content. Content (e.g., video content, audio content, multimedia content, etc.) may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). At **610** a plurality of portions of content from one or more content sources (e.g., the content source(s) **127**, etc.) may be determined. A device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may determine the plurality of portions of the content. The plurality of portions of the content may be based on a determined amount of time (e.g., time window, time duration, etc.). For example, each portion of the plurality of portions of the content may be based on and/or associated with an interval (e.g., 2-second interval, etc.) of the content. The interval may comprise and/or be associated with a frame (e.g., a portion) of the content, a group of frames of the content, a block of the content, or the like.

(104) At **620**, a plurality of content elements associated with the content may be determined. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine the plurality of content elements. Each portion of the plurality of portions of the content may be associated (e.g., tagged, etc.) with a respective content element of a plurality of content elements.

(105) A content element may be associated with an encoding parameter associated with the respective portion of the content. The encoding parameter may indicate a level of complexity (e.g., low complexity, medium complexity, high complexity, etc.) to encode (e.g., for the content management device **128** and/or any other device to encode) the respective portion of the content. The level of complexity may be based on one or more of a prediction frame error associated with the respective portion of the content, a relative size of an inter-coded frame of the respective portion of the content with respect to an intra-coded frame of the respective portion of the content, or a number of motion vectors to encode a frame of the respective portion of the content. The level of complexity may be based on a numerical count, an average, a categorization, or the like, of information about encoding a portion of the content. The level of complexity may be based on and/or determined by an amount of change between a plurality of pixels associated with a previous portion of the content and a corresponding plurality of pixels associated with the respective portion of the content. For example, a level of complexity to encode a portion of content may be low if an amount of change in pixels between the portions of content does not satisfy a threshold, or the level of complexity to encode the portion of content may be high if the amount of change in pixels between the portions of content satisfies or exceeds the threshold.

(106) A content element may be associated with attributes of content (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content. For example, the content element may be and/or be associated with metadata that indicates and/or causes a scene transition between a previous portion (or a post portion) of content and the respective portion of the content. As another example, the content element may be and/or be associated with metadata that associates the respective portion of the content with one or more advertisements. As another example, the content element may be and/or be associated with metadata that associates an audio level with the

respective portion of the content. A content element of a plurality of content elements may be determined and associated with each portion of a plurality of portions of the content based on any aspect of the content or a respective portion of the content.

(107) At **630**, a bitrate and/or a plurality of bitrates may be determined and associated with each portion of the plurality of portions of the content. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine, for each portion of the plurality of portions of the content, based on the respective content element, a respective bitrate of a plurality of bitrates. The plurality of content elements may be used to determine a plurality of bitrates to associate with the plurality of portions of content such that each portion of the plurality of portions of the content is associated with a content element and one or more bitrates, such as a low bitrate, a medium bitrate, a high bitrate, or the like. Any range of bitrates may be determined to be a low bitrate, a medium bitrate, or a high bitrate. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine that any bitrate is a low bitrate, a medium bitrate, or a high bitrate. Any range of bitrates may be determined to be a low bitrate, a medium bitrate, or a high bitrate by and/or based on any method, such as based on network (e.g., the network **116**, etc.) conditions (e.g., latency in a portion/channel of the network, network congestion, provisioned information and/or data transfer rates, etc.), user determined bitrate classifications (e.g., user defined bitrate ranges from high to low, etc.), dynamically determined bitrate classifications, or the like. Any bitrate and/or plurality of bitrates may be associated with any portion of the content based on a respective content element associated with the portion of the content.

(108) A bitrate and/or a plurality of bitrates may be associated with a portion of the plurality of portions of the content based on an associated level of complexity to encode the portion of the content (as indicated by a respective content element). The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine the bitrate and/or the plurality of bitrates and associate the bitrate and/or the plurality of bitrates with the portion of the plurality of portions of the content based on the associated level of complexity to encode the portion of the content (as indicated by the respective content element). For example, an impact on a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of a portion of content may be caused by network congestion (or any other cause) and may result in image artifacts, blurred images, illegible sound, paused/frozen content, and the like associated with the portion of content.

(109) A low level of complexity to encode a portion of content may indicate that the portion of the content may be provided to a network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content by the network device (a user of the network device) will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine that an encoding parameter associated/indicated by a content element indicates that a portion of content is associated with a level of complexity (e.g., a low level of complexity, a medium level of complexity, a high level of complexity, etc.) to encode (e.g., encoded by the encoding module **130** and/or any other device) the portion of the content. A low bitrate and/or a plurality of low bitrates may be associated with a portion of the content associated with a content element (e.g., an encoding parameter) that indicates a low level of complexity to encode the portion of the content. A high level of complexity to encode a portion of content may indicate that the portion of the content should be provided to the network device and/or any other device at a high bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content by the network device (a user of the network device) and/or any other device will be impacted if the portion of the content is not provided and/or consumed at a

high bitrate. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a high bitrate and/or a plurality of high bitrates to a portion of the content associated with a content element (e.g., encoding parameter) that indicates a high level of complexity to encode the portion of the content. Any bitrate and/or plurality of bitrates may be associated with any portion of content based on a content element associated with the portion of the content.

(110) A bitrate and/or a plurality of bitrates may be associated with each portion of the plurality of portions of the content based on attributes (e.g., a scene transition, a scene change, etc.) of the content associated with the respective portion of the content and indicated by a content element. For example, the device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a low bitrate and/or a plurality of low bitrates to a portion of the content associated with a content element that indicates a scene transition (e.g. scene change, etc.) associated with the portion(s) of the content. A quality of consumption associated with the portion of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate because a scene transition may be associated with a visually black screen, a lack of dialog/audio, or the like occurring at the point (e.g., portion of the content) of the scene transition. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a medium/high bitrate and/or a plurality of medium/high bitrates with portions of the content associated with a content element that indicates a scene transition (e.g. scene change, etc.) associated with the portion(s) of the content does not occur. A scene transition that does not occur during a portion of the content may be an indication (e.g., indicated by an associated content element, etc.) that the portion of the content is associated with an important scene of the content, such as a scene/portion of the content where a quality of consumption associated with the scene/portion will be perceived as impacted if the scene/portion of the content is consumed at a low bitrate. Any bitrate and/or plurality of bitrates may be associated with any portion of content based on a content element associated with the portion of the content.

(111) The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a bitrate and/or a plurality of bitrates to each portion of a plurality of portions of content based on additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with a respective portion of the content and indicated by a content element. For example, a bitrate (e.g., a low bitrate, a medium bitrate, a high bitrate, etc.) and/or a plurality of bitrates may be associated with a portion of content associated with a content element that indicates that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portion of the content. The advertisement may be associated with the portion of the content based on metadata associated with the advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) and/or the portion of the content. Any bitrate and/or any plurality of bitrates may be associated with portions of content associated with a content element that indicates that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portion of the content.

(112) The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may store the content and/or provide the content to storage (e.g., a database, a cloud-based storage, etc.). The device or any other device may store the content and/or provide the content to storage such that each portion of the content is stored with an association to a bitrate and/or a plurality of bitrates determined based on the content element associated with the respective portion of the content.

(113) At **640** the content may be provided to a network device. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may provide the content to the network device. The device may provide the content to the

network device based on a request for the content. The device may provide the content to the network device such that each portion of the content is provided at a bitrate and/or a plurality of bitrates determined based on the content element associated with the respective portion of the content. The content may be provided to the network device and/or any other device as streaming content, downloaded content, referenced and retrieved content, or the like. The content may be provided in a form comprising an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like. For example, the network device and/or any other device may receive/retrieve the content as a manifest file comprising a plurality of references to the content. The network device and/or any other device may receive/retrieve the content as a manifest file comprising a plurality of references to the content such that each portion of the plurality of portions of the content is associated with a bitrate and/or a plurality of bitrates determined based on a respective content element associated with a respective portion of the content. The content may be provided, retrieved, and/or received by the network device and/or any other device by any method.

(114) The network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) and/or any other device may consume the content and/or portions of the content in a best possible quality (e.g., based on a best possible bitrate, etc.) according to a plurality of content elements associated with the content (e.g., a plurality of portions of the content). The best possible quality may be a quality (e.g., picture quality, sound quality, visual quality, etc.) of the content with image artifacts, blurred images, illegible sound, paused/frozen content, or the like reduced and/or negligible.

(115) FIG. 7 is a flowchart of an example method **700** for providing variable bitrate content. At **710** content (e.g., video content, audio content, multimedia content, etc.) may be requested. The content may be requested by a network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) from one or more content sources (e.g., the content source(s) **127**, etc.). The content may comprise a plurality of portions (e.g., frames, segments, fragments, etc.). Each portion of the plurality of portions of the content may be based on and/or associated with an interval (e.g., 2-second interval, etc.) of the content. The interval may comprise and/or be associated with a frame (e.g., a portion) of the content, a group of frames of the content, a block of the content, or the like.

(116) At **720**, a plurality of content elements and a plurality of references to the content may be received. The network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) may receive the plurality of content elements and the plurality of references to the content. For example, network device may receive/retrieve a manifest file, an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like comprising the plurality of content elements and the plurality of references to the content.

(117) Each content element of the plurality of content elements may be associated with a portion of the plurality of portions of the content. A device (e.g., the content management device **128**, a computing device, a network device, a communication terminal, etc.) may determine the plurality of content elements. Each portion of the plurality of portions of the content may be associated (e.g., tagged, etc.) with a respective content element of a plurality of content elements.

(118) A content element may be associated with an encoding parameter associated with the respective portion of the content. The encoding parameter may indicate a level of complexity (e.g., low complexity, medium complexity, high complexity, etc.) to encode (e.g., for the content management device **128** and/or any other device to encode) the respective portion of the content. The level of complexity may be based on one or more of a prediction frame error associated with the respective portion of the content, a relative size of an inter-coded frame of the respective portion of the content with respect to an intra-coded frame of the respective portion of the content, or a number of motion vectors to encode a frame of the respective portion of the content. The level

of complexity may be based on a numerical count, an average, a categorization, or the like, of information about encoding a portion of the content. The level of complexity may be based on and/or determined by an amount of change between a plurality of pixels associated with a previous portion of the content and a corresponding plurality of pixels associated with the respective portion of the content. For example, a level of complexity to encode a portion of content may be low if an amount of change in pixels between the portions of content does not satisfy a threshold, or the level of complexity to encode the portion of content may be high if the amount of change in pixels between the portions of content satisfies or exceeds the threshold.

(119) A content element may be associated with attributes of content (e.g., a scene transition, scene change, etc.) associated with the respective portion of the content. For example, the content element may be and/or be associated with metadata that indicates and/or causes a scene transition between a previous portion (or a post portion) of content and the respective portion of the content. As another example, the content element may be and/or be associated with metadata that associates the respective portion of the content with one or more advertisements. As another example, the content element may be and/or be associated with metadata that associates an audio level with the respective portion of the content. A content element of a plurality of content elements may be determined and associated with each portion of a plurality of portions of the content based on any aspect of the content or a respective portion of the content.

(120) Each reference of the plurality of references may be associated with a portion of the plurality of portions of the content and one or more bitrates of a plurality of bitrates. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine, for each portion of the plurality of portions of the content, based on the respective content element, a respective bitrate of a plurality of bitrates. The plurality of content elements may be used to determine a plurality of bitrates to associate with the plurality of portions of content such that each portion of the plurality of portions of the content is associated with a content element and one or more bitrates, such as a low bitrate, a medium bitrate, a high bitrate, or the like. Any range of bitrates may be determined to be a low bitrate, a medium bitrate, or a high bitrate. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine that any bitrate is a low bitrate, a medium bitrate, or a high bitrate. Any range of bitrates may be determined to be a low bitrate, a medium bitrate, or a high bitrate by and/or based on any method, such as based on network (e.g., the network **116**, etc.) conditions (e.g., latency in a portion/channel of the network, network congestion, provisioned information and/or data transfer rates, etc.), user determined bitrate classifications (e.g., user defined bitrate ranges from high to low, etc.), dynamically determined bitrate classifications, or the like. Any bitrate and/or plurality of bitrates may be associated with any portion of the content based on a respective content element associated with the portion of the content.

(121) A bitrate and/or a plurality of bitrates may be associated with a portion of the plurality of portions of the content based on an associated level of complexity to encode the portion of the content (as indicated by a respective content element). The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine the bitrate and/or the plurality of bitrates and associate the bitrate and/or the plurality of bitrates with the portion of the plurality of portions of the content based on the associated level of complexity to encode the portion of the content (as indicated by the respective content element). For example, an impact on a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of a portion of content may be caused by network congestion (or any other cause) and may result in image artifacts, blurred images, illegible sound, paused/frozen content, and the like associated with the portion of content.

(122) A low level of complexity to encode a portion of content may indicate that the portion of the content may be provided to the network device (e.g., a content player, the media device **120**, a

smart device, the mobile device **124**, a computing device, etc.) at a low bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content by the network device (a user of the network device) will not be impacted (e.g., will not be substantially impacted, will be minimally impacted, etc.) by the low bitrate. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may determine that an encoding parameter associated/indicated by a content element indicates that a portion of content is associated with a level of complexity (e.g., a low level of complexity, a medium level of complexity, a high level of complexity, etc.) to encode (e.g., encoded by the encoding module **130** and/or any other device) the portion of the content. A low bitrate and/or a plurality of low bitrates may be associated with a portion of the content associated with a content element (e.g., an encoding parameter) that indicates a low level of complexity to encode the portion of the content. A high level of complexity to encode a portion of content may indicate that the portion of the content should be provided to the network device and/or any other device at a high bitrate because a quality of consumption (e.g., picture quality, sound quality, visual quality, etc.) of the portion of content by the network device (a user of the network device) and/or any other device will be impacted if the portion of the content is not provided and/or consumed at a high bitrate. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a high bitrate and/or a plurality of high bitrates to a portion of the content associated with a content element (e.g., encoding parameter) that indicates a high level of complexity to encode the portion of the content. Any bitrate and/or plurality of bitrates may be associated with any portion of content based on a content element associated with the portion of the content.

(123) A bitrate and/or a plurality of bitrates may be associated with each portion of the plurality of portions of the content based on attributes (e.g., a scene transition, a scene change, etc.) of the content associated with the respective portion of the content and indicated by a content element. For example, the device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a low bitrate and/or a plurality of low bitrates to a portion of the content associated with a content element that indicates a scene transition (e.g. scene change, etc.) associated with the portion(s) of the content. A quality of consumption associated with the portion of the content may not (will not) be perceived as impacted (e.g., visually/audibly detected by a user consuming the portion of the content) by a low bitrate because a scene transition may be associated with a visually black screen, a lack of dialog/audio, or the like occurring at the point (e.g., portion of the content) of the scene transition. The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a medium/high bitrate and/or a plurality of medium/high bitrates with portions of the content associated with a content element that indicates a scene transition (e.g. scene change, etc.) associated with the portion(s) of the content does not occur. A scene transition that does not occur during a portion of the content may be an indication (e.g., indicated by an associated content element, etc.) that the portion of the content is associated with an important scene of the content, such as a scene/portion of the content where a quality of consumption associated with the scene/portion will be perceived as impacted if the scene/portion of the content is consumed at a low bitrate. Any bitrate and/or plurality of bitrates may be associated with any portion of content based on a content element associated with the portion of the content.

(124) The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may associate a bitrate and/or a plurality of bitrates to each portion of a plurality of portions of content based on additional content related items (e.g., one or more advertisements, one or more references to related content, etc.) associated with a respective portion of the content and indicated by a content element. For example, a bitrate (e.g., a low bitrate, a medium bitrate, a high bitrate, etc.) and/or a plurality of bitrates may be associated with a portion of content associated with a content element that indicates that an advertisement (e.g., a portion(s)

of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portion of the content. The advertisement may be associated with the portion of the content based on metadata associated with the advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) and/or the portion of the content. Any bitrate and/or any plurality of bitrates may be associated with portions of content associated with a content element that indicates that an advertisement (e.g., a portion(s) of an advertisement, an advertisement segment, an advertisement spot, etc.) is associated with the portion of the content. (125) The device (e.g., the content management device **128**, the computing device, the network device, the communication terminal, etc.) may store the content and/or provide the content to storage (e.g., a database, a cloud-based storage, etc.). The device or any other device may store the content and/or provide the content to storage such that each portion of the content is stored with an association to a bitrate and/or a plurality of bitrates determined based on the content element associated with the respective portion of the content. The network device may receive/retrieve the content from storage and/or any other device. The network device may receive/retrieve the plurality of content elements and/or the content from storage and/or any other device based on the plurality of references to the content. For example, network device may receive/retrieve a manifest file, an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like comprising the plurality of content elements and the plurality of references to the content.

(126) At **730** one or more bitrates associated with the content may be selected. The network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) may select, based on a respective content element of the plurality of content elements, for each portion of the plurality of portions of the content, the one or more bitrates of the plurality of bitrates that are associated with the portion of the content. The network device may select the one or more bitrates according a content element associated with the portion of the content. The network device may select the one or more bitrates to achieve a best possible quality consumption of the portion of the content. For example, quality (e.g., picture quality, sound quality, visual quality, etc.) consumption of content may be content that may be consumed (e.g., displayed, watched, listened to, etc.) without perceived changes in video quality (e.g., no image artifacts, no blurred images, no unintentionally paused and/or delayed video, etc.), audio quality (e.g., no unintentional changes in volume, no changes in sound clarity, etc.), and or the like.

(127) The network device (e.g., a content player, the media device **120**, a smart device, the mobile device **124**, a computing device, etc.) may use the content element associated with the portion of the content to determine/select a bitrate of the one or more plurality of bitrates associated with the portion of the content that will enable the portion content to be consumed at the best quality. For example, if the network device (or a user associated with the network device) is consuming the content, the network device may automatically select/adjust a bitrate of the one or more bitrates associated with the portion of the content because the content element indicates that the quality of consumption will either not be impacted or will be enhanced by a selection/adjustment of a bitrate, such as an selection/adjustment to a lower bitrate, or a higher bitrate compared to a bitrate at which other portions of the content are/can be consumed.

(128) At **710** each portion of the plurality of portions of the content may be received/retrieved. The network device may receive/retrieve each portion of the plurality of portions of the content according to the selected bitrate. For example, the network device may use a manifest file, an index file, a multimedia file, a text file, code, a script, metadata, resource location(s), combinations thereof, or the like comprising the plurality of content elements and the plurality of references to the content to retrieve the content.

(129) FIG. **8** shows a system **800** for providing variable bitrate content. The server **110**, the application server **126**, the content source **127**, or the edge device **128** of FIG. **1**, or any other device/component described herein may be a computer **801** as shown in FIG. **8**.

(130) The computer **801** may comprise one or more processors **803**, a system memory **812**, and a bus **813** that couples various components of the computer **801** including the one or more processors **803** to the system memory **812**. In the case of multiple processors **803**, the computer **801** may utilize parallel computing.

(131) The bus **813** may comprise one or more of several possible types of bus structures, such as a memory bus, memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures.

(132) The computer **801** may operate on and/or comprise a variety of computer readable media (e.g., non-transitory). Computer readable media may be any available media that is accessible by the computer **801** and comprises, non-transitory, volatile and/or non-volatile media, removable and non-removable media. The system memory **812** has computer readable media in the form of volatile memory, such as random access memory (RAM), and/or non-volatile memory, such as read only memory (ROM). The system memory **812** may store data such as bitrate determination data **807** and/or program modules such as operating system **805** and content analysis software **806** that are accessible to and/or are operated on by the one or more processors **803**.

(133) The computer **801** may also comprise other removable/non-removable, volatile/non-volatile computer storage media. The mass storage device **804** may provide non-volatile storage of computer code, computer readable instructions, data structures, program modules, and other data for the computer **801**. The mass storage device **804** may be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.

(134) Any number of program modules may be stored on the mass storage device **804**. An operating system **805** and content analysis software **806** may be stored on the mass storage device **804**. One or more of the operating system **805** and content analysis software **806** (or some combination thereof) may comprise program modules and the content analysis software **806**. Bitrate determination data **807** may also be stored on the mass storage device **804**. Bitrate determination data **807** may be stored in any of one or more databases known in the art. The databases may be centralized or distributed across multiple locations within the network **815**.

(135) A user may enter commands and information into the computer **801** via an input device (not shown). Such input devices comprise, but are not limited to, a keyboard, pointing device (e.g., a computer mouse, remote control), a microphone, a joystick, a scanner, tactile input devices such as gloves, and other body coverings, motion sensor, and the like. These and other input devices may be connected to the one or more processors **803** via a human machine interface **802** that is coupled to the bus **813**, but may be connected by other interface and bus structures, such as a parallel port, game port, an IEEE 1394 Port (also known as a Firewire port), a serial port, network adapter **808**, and/or a universal serial bus (USB).

(136) A display device **811** may also be connected to the bus **813** via an interface, such as a display adapter **809**. It is contemplated that the computer **801** may have more than one display adapter **809** and the computer **801** may have more than one display device **811**. A display device **811** may be a monitor, an LCD (Liquid Crystal Display), light emitting diode (LED) display, television, smart lens, smart glass, and/or a projector. In addition to the display device **811**, other output peripheral devices may comprise components such as speakers (not shown) and a printer (not shown) which may be connected to the computer **801** via Input/Output Interface **810**. Any step and/or result of the methods may be output (or caused to be output) in any form to an output device. Such output may be any form of visual representation, including, but not limited to, textual, graphical, animation, audio, tactile, and the like. The display **811** and computer **801** may be part of one device, or separate devices.

(137) The computer **801** may operate in a networked environment using logical connections to one

or more remote computing devices **814a,b,c**. A remote computing device **814a,b,c** may be a personal computer, computing station (e.g., workstation), portable computer (e.g., laptop, mobile phone, tablet device), smart device (e.g., smartphone, smart watch, activity tracker, smart apparel, smart accessory), security and/or monitoring device, a server, a router, a network computer, a peer device, edge device or other common network node, and so on. Logical connections between the computer **801** and a remote computing device **814a,b,c** may be made via a network **815**, such as a local area network (LAN) and/or a general wide area network (WAN). Such network connections may be through a network adapter **808**. A network adapter **808** may be implemented in both wired and wireless environments. Such networking environments are conventional and commonplace in dwellings, offices, enterprise-wide computer networks, intranets, and the Internet.

(138) Application programs and other executable program components such as the operating system **805** are shown herein as discrete blocks, although it is recognized that such programs and components may reside at various times in different storage components of the computing device **801**, and are executed by the one or more processors **803** of the computer **801**. An implementation of content analysis software **806** may be stored on or sent across some form of computer readable media. Any of the disclosed methods may be performed by processor-executable instructions embodied on computer readable media.

(139) While specific configurations have been described, it is not intended that the scope be limited to the particular configurations set forth, as the configurations herein are intended in all respects to be possible configurations rather than restrictive.

(140) Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method claim does not actually recite an order to be followed by its steps or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is no way intended that an order be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps or operational flow; plain meaning derived from grammatical organization or punctuation; the number or type of configurations described in the specification.

(141) It will be apparent to those skilled in the art that various modifications and variations may be made without departing from the scope or spirit. Other configurations will be apparent to those skilled in the art from consideration of the specification and practice described herein. It is intended that the specification and described configurations be considered as exemplary only, with a true scope and spirit being indicated by the following claims.

Claims

1. A method comprising: determining, for a first portion of a plurality of portions of content, a level of encoding complexity, wherein the first portion follows an adjacent second portion of the plurality of portions of the content; determining, based on the level of encoding complexity, an output bitrate to assign to the first portion; determining, based on an attribute of metadata of the first portion, a scene change between the adjacent second portion and the first portion; based on the determination of the scene change between the adjacent second portion and the first portion and based on the output bitrate of the first portion being less than an output bitrate of the adjacent second portion, adjusting the output bitrate of the first portion to be at least as great as the output bitrate of the adjacent second portion; and sending, based on a request for the content, the first portion at the output bitrate of the first portion.
2. The method of claim 1, wherein determining the level of encoding complexity comprises: determining, based on a content element associated with the first portion, the level of encoding complexity, wherein the content element is indicative of the level of encoding complexity.
3. The method of claim 2, wherein the content element comprises an indication of at least one of: a

- prediction frame error; a size of an inter-coded frame with respect to an intra-coded frame; a quantity of motion vectors; or an amount of change between a plurality of pixels associated with the first portion and a plurality of pixels associated with a previous portion.
4. The method of claim 2, wherein the content element comprises the metadata indicative of at least one of: at least one advertisement; or an audio level.
 5. The method of claim 1, wherein determining the output bitrate to assign to the first portion comprises: determining that the level of encoding complexity associated with the first portion satisfies a threshold; and determining, based on the level of encoding complexity associated with the first portion satisfying the threshold, the output bitrate to assign to the first portion.
 6. The method of claim 5, further comprising: determining, for at least one further portion of the plurality of portions of the content, that a level of encoding complexity associated with the at least one further portion does not satisfy the threshold; and determining, based on the level of encoding complexity associated with the at least one further portion not satisfying the threshold, an output bitrate to assign to the at least one further portion.
 7. The method of claim 6, wherein the output bitrate to assign to the at least one further portion comprises a lower bitrate than the output bitrate for the first portion.
 8. A method comprising: receiving, based on a request for content, a manifest file identifying: a plurality of portions of the content and a level of encoding complexity for each portion of the plurality of portions of the content; determining, based on the level of encoding complexity identified in the manifest file for a first portion of the plurality of portions of the content, an output bitrate to request for the first portion, wherein the first portion follows an adjacent second portion of the plurality of portions of the content; determining, based on an attribute of metadata of the first portion, a scene change between the adjacent second portion and the first portion; based on the determination of the scene change between the adjacent second portion and the first portion and based on the output bitrate of the first portion being less than an output bitrate of the adjacent second portion, adjusting the output bitrate of the first portion to be at least as great as the output bitrate of the adjacent second portion; and sending a request for the first portion at the output bitrate.
 9. The method of claim 8, wherein the manifest file comprises a content element associated with the first portion, and wherein the content element is indicative of the level of encoding complexity.
 10. The method of claim 9, wherein the content element comprises an indication of at least one of: a prediction frame error; a size of an inter-coded frame with respect to an intra-coded frame; a quantity of motion vectors; or an amount of change between a plurality of pixels associated with the first portion and a plurality of pixels associated with a previous portion.
 11. The method of claim 9, wherein the content element comprises the metadata indicative of at least one of: at least one advertisement; or an audio level.
 12. The method of claim 8, further comprising: determining that the level of encoding complexity associated with the first portion satisfies a threshold; and determining, based on the level of encoding complexity associated with the first portion satisfying the threshold, the output bitrate to request for the first portion.
 13. The method of claim 12, further comprising: determining, for at least one further portion of the plurality of portions of the content, that a level of encoding complexity associated with the at least one further portion does not satisfy the threshold; and determining, based on the level of encoding complexity associated with the at least one further portion not satisfying the threshold, an output bitrate to request for the at least one further portion.
 14. The method of claim 13, wherein the output bitrate for the at least one further portion comprises a lower bitrate than the output bitrate for the first portion.
 15. A method comprising: determining, for a first portion and a second portion of a plurality of portions of content, a level of encoding complexity; determining, based on the level of encoding complexity associated with the first portion of the plurality of portions of the content, a first output

bitrate of a plurality of output bitrates to assign to the first portion; determining, based on the level of encoding complexity associated with the second portion of the plurality of portions of the content, a second output bitrate of the plurality of output bitrates to assign to the second portion; determining, based on an attribute of metadata of an adjacent portion, of the plurality of portions of the content, that follows the first portion and precedes the second portion, a scene change between the adjacent portion and the second portion; based on the determination of the scene change between the adjacent portion and the second portion and based on the second output bitrate of the second portion being less than a third output bitrate of the adjacent portion, adjusting the second output bitrate of the second portion to be at least as great as the third output bitrate of the adjacent portion; and sending, based on a request for the content, the first portion at the first output bitrate and the second portion at the second output bitrate.

16. The method of claim 15, wherein the level of encoding complexity comprises at least one of: a prediction frame error; a encoding quantization level; a size of an inter-coded frame with respect to an intra-coded frame; a quantity of motion vectors; or a amount of pixel change.

17. The method of claim 15, wherein the second output bitrate comprises a lower output bitrate than the first output bitrate.

18. The method of claim 15, further comprising: generating a manifest file identifying the plurality of portions of the content and the level of encoding complexity for the first portion and the second portion of the plurality of portions of the content; and sending the manifest file.

19. The method of claim 15, wherein the first portion and the second portion of the plurality of portions of the content is associated with a content element indicative of the level of encoding complexity, and wherein determining, for the first portion and the second portion of the plurality of portions of the content, the level of encoding complexity comprises: determining, based on the content element, the level of encoding complexity.

20. The method of claim 19, wherein the content element for the first portion and the second portion of the plurality of portions of the content comprises an indication of at least one of: a prediction frame error; a size of an inter-coded frame with respect to an intra-coded frame; a quantity of motion vectors; or an amount of change between a plurality of pixels associated with the the first portion or the second portion and a plurality of pixels associated with a previous portion.
